

Sami Hatoum

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Experience

PricewaterhouseCoopers (PwC) | Software Consulting Intern

Jun 2024 – Aug 2024

- Built a **sentiment analysis model** using **PyTorch BERT** to automate customer feedback classification for a retail client. Achieved **91% F1 score** on an imbalanced dataset (3:1 negative to positive), processing **10,000 reviews daily**
- Engineered containerised deployment with **Flask REST API** on **AWS EC2** with auto-scaling across multiple instances. Validated **sub-500ms latency** and **150 requests/second** throughput under load testing with **Locust**
- Set up **CloudWatch** monitoring dashboards and wrote an automated retraining pipeline triggered when weekly evaluation accuracy dropped below **85%**.
- Optimised preprocessing with **pandas** vectorisation, reducing daily batch processing time by **60%**

PricewaterhouseCoopers (PwC) | Software Engineering Intern

Aug 2023 – Sep 2023

- Developed an internal **client onboarding dashboard** in **React** and **Node.js** that replaced a manual spreadsheet workflow for a financial services team, reducing onboarding processing time from **3 days to 4 hours** across 200+ users
- Designed and implemented a **PostgreSQL** data model with **role-based access control** across 4 user roles, integrating with PwC's internal **SSO authentication** system via **OAuth 2.0**
- Wrote **CI/CD pipelines** in **GitHub Actions** with automated testing (**Jest**, **Supertest**) and staged deployments to **AWS**, achieving **90%+ code coverage** across 45 unit and integration tests spanning the **REST API** and data access layers

Education

University College London | MSc Artificial Intelligence and Medical Imaging

Predicted Distinction

Modules: Statistical Modelling, Inverse Problems, Applied Deep Learning, Machine Learning in Medical Imaging

University of St Andrews | BSc Computer Science

First Class Honours, Dean's List

Modules: Computational Complexity, Constraint Programming, Logic & Reasoning, Databases, Data structures & Algorithms

Projects

Database Query Optimiser

- Developed a **PostgreSQL** analysis tool in **Python** that parses **query ASTs**, evaluates **EXPLAIN plans**, and recommends indexes. Reduced query execution time from **2.4s to 420ms** on table scans over **500K rows**
- Reviewed **200+ production queries** across a **five-million-row database** (12 tables), identifying **21 missing indexes** that reduced average query time by **65%** and eliminated most sequential scans
- Deployed on **AWS ECS** with **RDS PostgreSQL**, implementing **CloudWatch** alerts for query latency above 5s and **auto-scaling groups** for batch analysis traffic spikes

Selective Prediction for Medical Image Segmentation (MSc Dissertation)

- Built a **model-agnostic selective prediction framework** that determines when AI segmentation outputs are reliable or should defer to a human expert, reducing silent failures in clinical deployment of **nnU-Net** on pelvic MRI
- Implemented **Monte Carlo Dropout** and **Test-Time Augmentation** as independent uncertainty estimators, with entropy, mutual information, and coefficient of variation aggregation to produce per-scan deferral decisions
- Achieved **0.89 Dice score at 85% coverage** on 9-class pelvic segmentation, a **+0.02 Dice improvement** over accepting all predictions at full coverage. Evaluation pipeline generates accuracy-coverage curves, calibration plots, and per-class reliability diagrams across all folds

Hierarchically Weighted Image Segmentation (Group Project)

- Designed a **custom hierarchical loss function** with domain-specific penalty weights and asymmetric error costs, prioritising high-risk regions where misclassification has severe downstream consequences
- Trained a **3D Residual Encoder U-Net** on 589 multi-institutional MRI volumes achieving **0.886 mean Dice score**; improved segmentation of the smallest and hardest class from **0.630 to 0.659** over unweighted baseline
- Ran **ablation studies** across 3 loss functions and 3 weighting schemes; found that radiation sensitivity weighting alone accounted for **70%+** of the Dice gain on under-segmented organs, informing final model selection

Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL

Frameworks: React, Next.js, Node.js, Express.js, Flask, React Native, REST APIs

Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Jira, Agile, Linux

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, NLP, U-Net, BERT, Pandas, NumPy